Sri SAI RAM ENGINEERING COLLEGE | IEEE Reliability Society | IEEE Madras Section

TRACK 1

# INTELLECT HACK 2026

## KEYSTONE

Autonomous Structural Monitoring & Digital Preservation of Heritage Infrastructure

*Measure it before we lose it.*


**Team NeuraCoders**

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INTELLECT HACK 2026
AI HACKATHON

INNOVATE. BULID. DEMONSTRATE.

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# PROBLEM STATEMENT

## India protects 3,698 national monuments by looking at them.

### HOW A MONUMENT IS CHECKED TODAY

**VISIT**
A conservator travels to the site. One officer may cover several monuments.

>

**LOOK**
Damage is judged by eye and described in words — "minor cracking".

>

**FILE**
The note goes into a circle office. Nobody aggregates or trends it.

**Not one number is recorded.** So next year's visit has nothing to compare against, and slow movement is never detected — only its eventual result.

### WHAT THE AUDITORS FOUND

| | |
|---|---|
| 3,698 | centrally protected monuments, plus tens of thousands state-held |
| 0 | complete physical surveys of them since Independence |
| 92 | untraceable at CAG audit, from 1,655 actually inspected |
| 546 | encroachments found, where ASI had reported 249 |

*Source: CAG Performance Audit 18/2013 and follow-up 10/2022. Laser scanning has reached Nalanda, Lingaraja Temple and one Karnataka programme — the flagships. Everything else gets the clipboard.*

### WHAT THAT COSTS

**Damage stays invisible until it is advanced**

A wall can lean four centimetres across a year and look identical to the eye. Settlement, lean and crack growth are all slow. No inspector can hold last year's wall in their head, so decay is noticed only once it is obvious — which is to say, late.

**Condition is an opinion, not a number**

Two officers describe the same wall differently, and neither description can be set against last year's. Nothing is auditable. Which of a circle's hundred monuments most needs money this year is a judgement call with no evidence behind it.

**If it falls, the record falls with it**

A photograph has no scale: you cannot tell from it whether a column is three metres or four. So when a monument is damaged there is nothing dimensional to rebuild from, and restoration becomes educated guesswork.

# PROPOSED SOLUTION

*A monument measures itself, repeatedly, from a phone or a sub-₹1-lakh drone — and raises its own alarm.*

**CAPTURE**
drone / phone imagery

>

**RECONSTRUCT**
scaled 3D point cloud

>

**ARCHIVE**
permanent measured record

>

**RE-CAPTURE**
same site, months later

>

**DIFFERENCE**
M3C2 change detection

>

**ACT**
inspection order / access restriction

*Steps 4-6 repeat on every capture cycle. The system decides; the conservator validates.*

## PRESERVATION

*A permanent, dimensionally accurate record where none exists today.*

- Every visible element measured at true scale
- Plans, elevations and sections exportable for restoration
- Survives the monument — loss stops being total loss
- VR and web access for students and researchers

## PROTECTION

*Autonomous detection of structural change between capture cycles.*

- Millions of comparison points, not twenty tape readings
- Objective and repeatable across inspectors and years
- Severity classification triggers action without a human noticing
- Prioritises which monuments a short-staffed circle visits first

# TECHNICAL APPROACH

## Photograph it. Measure it. Photograph it again. Let the computer find what moved.

*Every tool below is open-source and established. The contribution is assembling them at consumer cost, repeating the capture, and ending in an action.*

### 01 CAPTURE

*photographs in*

- Several hundred overlapping photos of the monument, from a smartphone or a ₹35,000 drone
- Coverage includes the rear walls and low courses that visitor photographs never capture
- Surveyed markers of known length placed in shot — *this is what gives the model real-world dimensions*

### 02 RECONSTRUCT

*measured 3D model out*

- COLMAP works out where each photo was taken from, and builds a rough 3D shape
- OpenMVS fills that into a dense point cloud — millions of measured 3D points
- The model is then scaled to the survey markers — *this is what separates a measured record from a game asset*

### 03 COMPARE

*change detected*

- A later capture is aligned onto the first using the same fixed markers
- **M3C2** — the standard algorithm for comparing two point clouds — measures how far every point moved
- A trained crack-detection model measures how far cracks have widened since last time

### 04 THE SYSTEM ACTS ON ITS OWN

*No human has to notice. Crossing a threshold is what triggers the action.*

| | | |
|---|---|---|
| **INFORMATIONAL** | Movement too small to be sure it is real | Recorded in the monument's file, nothing more |
| **WATCH** | Real movement, but within normal seasonal range | Marked for the next visit; capture more often |
| **ALERT** | Movement beyond seasonal range, or a crack widening | Inspection order issued; site conservator notified |
| **CRITICAL** | Fast movement, or movement in a wall holding weight | Visitors kept out of the zone; structural engineer called |

**BUILT WITH** COLMAP · OpenMVS · Open3D · CloudCompare · PyTorch · GDAL · PostGIS · FastAPI · React · Potree  All open-source. No licence fees.

# FEASIBILITY & VIABILITY

*We are not competing with laser scanning. We are covering what it can never reach.*

## MANUAL INSPECTION

- ~20 tape measurements, once
- Subjective — varies by inspector
- Not comparable year to year
- No record if the monument is lost
- Cost: staff time only

## TERRESTRIAL LIDAR

- Millimetre accuracy — the gold standard
- Works in darkness, penetrates vegetation
- Needs a trained operator and days per site
- Too costly to repeat, so one-shot in practice
- Cost: ₹5-75 lakh a rig, plus operator and survey time

## KEYSTONE

- 1-3 cm with ground control points
- Millions of comparison points
- Cheap enough to repeat every cycle — the whole point
- Scales to the full national inventory
- Cost: sub-₹1L drone, or a phone already owned

## STATED LIMITS

- Accuracy floor is 1-3 cm — an order of magnitude below lidar, and we say so
- Surfaces only: no foundations, wall core or mortar chemistry
- Seasonal movement is a real confound, mitigated not eliminated
- Vegetation and restricted interiors reduce coverage — gaps are flagged, not silent

## WHY IT DEPLOYS

- A conservator already visits the site — capture fits inside that visit
- No new cadre, no procurement cycle, no scanner logistics
- Answers a standing parliamentary directive for digital condition logs
- Entire pipeline is open-source; no per-seat licensing

# IMPACT & BENEFITS

*What the measurements actually do — during the monument's life, and after it.*

## PREVENTION

Gross structural change is caught between inspection cycles instead of after failure. Monuments are load-bearing public spaces receiving thousands of visitors daily — failure is a mass-casualty event, not just a heritage loss.

## RECONSTRUCTION

The point cloud slices into plans, elevations and sections. Builders build from drawings; photographs never become drawings. Stone is then machine-carved to recorded geometry — as with Palmyra's Arch of Triumph, recreated in marble from crowdsourced photographs after its destruction.

## ANASTYLOSIS

After collapse there is rubble. Surviving fragments are geometrically matched back to their original positions — impossible without a measured record, and routine conservation practice with one.

## COVERAGE

Shifts heritage recording from flagship-only to nationally feasible by removing the cost barrier. The thousands of monuments that will never see a laser scanner finally get a baseline.

## ACCOUNTABILITY

CAG found 546 encroachments where ASI had reported 249. Measured records are auditable; written impressions are not. Condition becomes a number a circle office can be held to.

## EDUCATION & ACCESS

Students walk through monuments in VR from any classroom. Cross-sections show construction logic no photograph conveys. Researchers measure remotely instead of travelling.

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# PROGRESS AND PROTOTYPE

*One month to a working reconstruction and a demonstrated change detection.*

**WEEK 1**

## Concept & Deck

- Problem framing validated against CAG audit findings
- Architecture and autonomy loop finalised
- Accuracy limits scoped honestly
- Submission complete

**WEEK 2**

## Reconstruction Proof

- Capture one accessible local structure by smartphone
- COLMAP to dense cloud, scale-corrected against a measured bar
- Validate: tape a real dimension, report the actual error

**WEEK 3**

## Change Detection Proof

- Second capture of the same structure
- ICP registration, M3C2 differencing, visualised diff
- Introduce a known deliberate change to prove sensitivity

**WEEK 4**

## Autonomy & Demo

- Severity thresholds and action triggering
- Web viewer with point cloud rendering
- Three-minute demo rehearsed; hostile Q&A rehearsed

**DEMO SUCCESS CRITERION** A measured model of a real structure, a second capture, and an automatically flagged change with a triggered action — end to end, on screen, in under three minutes.

# CONCLUSION

*Every component exists. Nobody has assembled them at this price point for this use case.*

**THE GAP**

Laser scanning is the right tool for a hundred monuments. There are tens of thousands. ASI cannot reliably say how many it protects, and no full physical survey has been completed since Independence.

**THE SYSTEM**

Consumer-cost capture, dimensionally corrected, repeated over time, differenced automatically, and terminating in an action rather than a report a human may never read.

**THE HONEST CLAIM**

If the monument is lost, our record preserves its architecture completely — form, proportion, ornament, dimension. It cannot preserve what was inside the walls. A reconstruction guided by it would be faithful in every visible respect.

**Manual inspection tells you what a monument looks like today. It cannot tell you what changed since last year.**